Balance Table

Covariate balance and wave composition (companion to Causal_Forest.Rmd)

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We assess covariate balance between treatment and control groups in our restricted Type 0/1 sample to validate it before subsequent treatment-effect estimation using interaction regressions and causal forests. These checks are conducted independently of either estimation method and are therefore documented separately. Both analyses use the sample prepared by cleaned_data.Rmd and saved as cleaned data.Rda, so the cleaning script must be knitted first.

1 Setup and data preparation

```
required_packages <- c("estimatr", "dplyr", "knitr")
missing_packages  <- setdiff(required_packages, rownames(installed.packages()))
if (length(missing_packages) > 0) {
  install.packages(missing_packages, repos = "https://cloud.r-project.org")
}
library(estimatr)
library(dplyr)
library(knitr)
```

```
# cleaned_data.Rmd builds and saves the analysis sample (df_int1)
rda_path <- "cleaned data.Rda"
if (!file.exists(rda_path)) {
  stop("Cleaned data not found. Knit cleaned_data.Rmd first to create: ",
       normalizePath(rda_path, mustWork = FALSE))
}
loaded <- load(rda_path)
stopifnot("df_int1" %in% loaded)
```

```r
# Pre-treatment resume characteristics
all_pretreat_cov <- c(
  "language_skills",
  "female",
  "ba_quality",
  "extracurricular_skills",
  "ma",
  "same_exp",
  "exp_highquality"
)

# Missing values are NOT recoded in cleaned_data.Rmd: same_exp is NA for every resume in
# wave 2. The balance statistics below therefore use the observed resumes only, so the
# same_exp row covers waves 0-1.
# The wave is saved as a factor; use a numeric code (0, 1, 2).
df_int1 <- df_int1 %>%
  mutate(fall_data = as.numeric(as.character(fall_data)))

waves <- sort(unique(df_int1$fall_data))
```

2 Covariate balance table

To assess whether our sample restrictions (to Types 0 and 1) preserve comparability between treatment (D = 1, foreign-sounding name) and control (D = 0, English-sounding name) groups, we first examine the distribution of pre-treatment resume characteristics across both groups.

We calculate the Standardised Mean Difference (SMD) for each covariate:

$$SMD = \frac{\bar{X}_{D=1} - \bar{X}_{D=0}}{\sqrt{\frac{s^2_{D=1} + s^2_{D=0}}{2}}}$$

Covariates with $|SMD| \geq 0.1$ are flagged for further investigation. We also report p-values from individual regressions of each covariate on treatment assignment, with standard errors clustered at the firm level. Wave is a three-level variable (0, 1, 2), so each wave is represented by a separate binary indicator rather than treated as a continuous variable.

The N columns report the number of resumes with $X = 1$ in each treatment group, while the missing column records the number of missing observations. As `same_exp` is only observed in Waves 0-1, its SMD and p-value are calculated using those waves only.

```r
smd <- function(x, d) {
  x1 <- x[d == 1]
  x0 <- x[d == 0]
  (mean(x1, na.rm = TRUE) - mean(x0, na.rm = TRUE)) /
    sqrt((var(x1, na.rm = TRUE) + var(x0, na.rm = TRUE)) / 2)
}

# One dummy per wave
df_bal <- df_int1
for (k in waves) df_bal[[paste0("wave_", k)]] <- as.numeric(df_bal$fall_data == k)

bal_vars <- c(all_pretreat_cov, paste0("wave_", waves))
```

```r
balance_row <- function(v, dat) {
  x <- as.numeric(dat[[v]])
  p <- tryCatch({
    # lm_robust drops resumes where the variable is missing
    fit <- lm_robust(as.formula(paste(v, "~ D")), data = dat,
                     clusters = firmid, se_type = "CR2")
    unname(fit$p.value["D"])
  }, error = function(e) NA_real_)
  # Counts only make sense for 0/1 variables
  is_bin <- all(x[!is.na(x)] %in% c(0, 1))
  data.frame(variable = v,
             n_D0     = if (is_bin) sum(x[dat$D == 0] == 1, na.rm = TRUE) else NA_integer_,
             n_D1     = if (is_bin) sum(x[dat$D == 1] == 1, na.rm = TRUE) else NA_integer_,
             mean_D0  = mean(x[dat$D == 0], na.rm = TRUE),
             mean_D1  = mean(x[dat$D == 1], na.rm = TRUE),
             SMD      = smd(x, dat$D),
             p_clustered = p,
             n_missing = sum(is.na(x)))
}

bal_tab <- bind_rows(lapply(bal_vars, balance_row, dat = df_bal)) %>%
  mutate(flag = ifelse(abs(SMD) >= 0.1, "check", ""))


bal_tab_display <- bal_tab %>%
  mutate(p_clustered = ifelse(
    is.na(p_clustered),
    NA_character_,
    ifelse(p_clustered < 0.001,
           "<0.001",
           sprintf("%.3f", p_clustered))
  ))

kable(bal_tab_display, digits = 3,
      col.names = c("variable", "N D=0", "N D=1",
                    "mean D=0", "mean D=1",
                    "SMD", "p (clustered)", "missing", "flag"),
      caption = paste("Balance of resume characteristics by name type",
                      "(D = 1 foreign name)"))
```

Table 1: Balance of resume characteristics by name type (D = 1 foreign name)

variable	N D=0	N D=1	mean D=0	mean D=1	SMD	p (clustered)	missing	flag
language_skills	748	676	0.247	0.257	0.022	0.384	0	
female	1544	1332	0.510	0.506	-0.008	0.767	0	
ba_quality	1784	1673	0.590	0.636	0.095	<0.001	0	
extracurricular_skills	1817	1600	0.600	0.608	0.016	0.593	0	
ma	620	522	0.205	0.198	-0.016	0.547	0	
same_exp	859	639	0.433	0.452	0.037	0.250	2260	
exp_highquality	1001	836	0.331	0.318	-0.028	0.328	0	

variable	N D=0	N D=1	mean D=0	mean D=1	SMD	p (clustered)	missing	flag
wave_0	953	949	0.315	0.361	0.097	<0.001	0	
wave_1	1029	466	0.340	0.177	-0.379	<0.001	0	check
wave_2	1044	1216	0.345	0.462	0.241	<0.001	0	check

From table 1, most resume characteristics are comparable between treatment and control groups, with absolute SMDs below 0.1. However, ba_quality has a statistically significant difference ($p < 0.001$), although its SMD of 0.095 falls just below the 0.1 threshold.

Notably, more substantial imbalances are observed in wave composition, particularly Waves 1 and 2 (both $p < 0.001$). These differences suggest that treatment and control resumes are not distributed equally across experimental waves.

We therefore examine wave composition and reassess covariate balance after controlling for wave in the following section.

3 Wave checks

Table 1 shows substantial differences in treatment shares across waves. We conduct two additional checks to investigate whether these differences reflect changes in sample composition across waves or potential covariate imbalance within waves.

3.1 Wave composition

We first examine which name ethnicities are represented in each wave and the corresponding treatment shares.

Table 2 shows that treatment shares vary considerably across waves, ranging from 31.2% in Wave 1 to 53.8% in Wave 2. The ethnicity breakdown above the Table 2 shows that Wave 0 includes Chinese and Indian names, Wave 1 includes Pakistani names, and Wave 2 includes Chinese and Indian names again, alongside Canadian names in all three waves.

This suggests that the differences in treatment shares are associated with changes in the ethnic composition of the experimental sample across waves, rather than necessarily indicating a failure of randomization.

```
# Which name types appear in which wave?
print(table(wave = df_int1$fall_data,
            ethnicity = as.character(df_int1$name_ethnicity)))
```

```
##     ethnicity
## wave Canada Chinese Indian Pakistani
##    0    953     481    468         0
##    1   1029       0      0       466
##    2   1044     517    699         0
```

```
# Treatment share and callback rate by wave
wave_tab <- df_int1 %>%
  group_by(wave = fall_data) %>%
  summarise(n = n(),
            treated_share    = mean(D),
```

```
            callback_rate_D0 = mean(as.numeric(callback)[D == 0]),
            callback_rate_D1 = mean(as.numeric(callback)[D == 1]),
            .groups = "drop")

kable(wave_tab, digits = 3, caption = "Treatment share and callback rate by wave")
```

Table 2: Treatment share and callback rate by wave

wave	n	treated_share	callback_rate_D0	callback_rate_D1
0	1902	0.499	0.158	0.115
1	1495	0.312	0.127	0.079
2	2260	0.538	0.141	0.098

3.2 Balance within wave

We then examine if pre-treatment resume characteristics remain balanced between treatment and control groups after accounting for wave composition. We regress each covariate on treatment assignment and wave fixed effects, with standard errors clustered at the firm level.

For each covariate we also report a wave-adjusted standardised mean difference. Its numerator is the regression coefficient on D (the difference in means holding wave fixed). Its denominator is the *raw* pooled standard deviation, the same one used in Table 1:

$$SMD_{adj} = \frac{\hat{\beta}_D}{\sqrt{\frac{s^2_{D=1}+s^2_{D=0}}{2}}}$$

The denominator is deliberately not recomputed after adjustment. Keeping it fixed means the raw and adjusted SMDs are on the same scale, so any change between them reflects a change in the mean difference and not a change in the scaling. We retain |SMD| = 0.1 as a diagnostic threshold for assessing covariate balance.

Table 3 shows that none of the covariates exhibit statistically significant differences between treatment and control groups after controlling for wave, and that every adjusted SMD is below 0.1 in absolute value. Notably, the difference in ba_quality, which was statistically significant in Table 1, is no longer significant (p = 0.227), and its SMD falls from 0.095 to a much smaller value. Figure 1 shows the raw and adjusted SMDs together.

Overall, the results suggest that the differences observed in Table 1 are largely associated with differences in wave composition. We find no statistically significant evidence of covariate imbalance within waves, consistent with the original experimental randomization.

Note: For each covariate, we estimate the regression

$$X_i = \alpha + \beta D_i + \delta_{w(i)} + \varepsilon_i$$

. Adjusted means are obtained from predicted covariate values for D = 0 and D = 1, controlling for waves, while adjusted differences correspond to the estimated coefficient on D. The adjusted SMD divides this coefficient by the raw pooled standard deviation; the raw SMD is the unadjusted standardised difference from Table 1, computed on the same observations. The regression coefficient weights the within-wave differences by precision, not by wave sample share. Standard errors are clustered at the firm level. The `same_exp` regression is restricted to Waves 0-1 due to structural missingness in Wave 2. Values are rounded to three decimal places.

```r
# Using regression, get wave-adjusted means, differences and SMDs

within_row_adj <- function(v) {
  dat <- df_int1
  dat$x <- as.numeric(dat[[v]])

  # Keep only observations used in the regression
  dat <- dat[!is.na(dat$x) &
               !is.na(dat$D) &
               !is.na(dat$fall_data) &
               !is.na(dat$firmid), ]

  # Same regression as original Table 3
  fit <- lm_robust(
    x ~ D + factor(fall_data),
    data = dat,
    clusters = firmid,
    se_type = "CR2"
  )

  # Predict for D=0 and D=1 controlling for wave
  dat0 <- dat
  dat0$D <- 0

  dat1 <- dat
  dat1$D <- 1

  mean_D0_adj <- mean(predict(fit, newdata = dat0))
  mean_D1_adj <- mean(predict(fit, newdata = dat1))

  # Raw pooled SD (unadjusted), the same denominator as the Table 1 SMD
  x1 <- dat$x[dat$D == 1]
  x0 <- dat$x[dat$D == 0]
  sd_pooled <- sqrt((var(x1) + var(x0)) / 2)

  b <- unname(coef(fit)["D"])

  data.frame(
    variable = v,
    mean_D0_adj = mean_D0_adj,
    mean_D1_adj = mean_D1_adj,
    diff_D1_minus_D0 = b,
    se = unname(fit$std.error["D"]),
    SMD_raw = (mean(x1) - mean(x0)) / sd_pooled,
    SMD_adj = b / sd_pooled,
    p_clustered = unname(fit$p.value["D"])
  )
}

# Generate within-wave table
within_tab_adj <- bind_rows(
  lapply(all_pretreat_cov, within_row_adj)
)
```

```r
# Format p-values for display
within_tab_adj_display <- within_tab_adj %>%
  mutate(
    p_clustered = ifelse(
      is.na(p_clustered),
      NA_character_,
      ifelse(
        p_clustered < 0.001,
        "<0.001",
        sprintf("%.3f", p_clustered)
      )
    )
  )

# Display Table 3
knitr::kable(
  within_tab_adj_display,
  digits = 3,
  col.names = c(
    "Covariate",
    "Adjusted Mean D=0",
    "Adjusted Mean D=1",
    "Adjusted Difference",
    "SE",
    "SMD (raw)",
    "SMD (adjusted)",
    "p (clustered)"
  ),
  caption = paste(
    "Balance within wave -",
    "wave-adjusted means, differences and SMDs",
    "(firm-clustered)"
  )
)
```

Table 3: Balance within wave - wave-adjusted means, differences and SMDs (firm-clustered)

Covariate	Adjusted Mean D=0	Adjusted Mean D=1	Adjusted Difference	SE	SMD (raw)	SMD (adjusted)	p (clustered)
language_skills	0.253	0.250	-0.003	0.011	0.022	-0.006	0.811
female	0.509	0.508	-0.001	0.014	-0.008	-0.001	0.963
ba_quality	0.604	0.619	0.015	0.012	0.095	0.030	0.227
extracurricular_skills	0.601	0.608	0.007	0.015	0.016	0.015	0.609
ma	0.206	0.198	-0.008	0.011	-0.016	-0.020	0.468
same_exp	0.441	0.441	0.000	0.016	0.037	0.001	0.977
exp_highquality	0.320	0.330	0.010	0.014	-0.028	0.021	0.473

```r
# Love plot: raw vs wave-adjusted |SMD| for each resume characteristic
lp <- within_tab_adj[order(abs(within_tab_adj$SMD_raw)), ]
y  <- seq_len(nrow(lp))
```

```r
xmax <- max(0.12, abs(lp$SMD_raw), abs(lp$SMD_adj)) * 1.1

op <- par(mar = c(4, 11, 1, 1))
plot(abs(lp$SMD_raw), y, xlim = c(0, xmax), ylim = c(0.5, length(y) + 0.5),
     pch = 16, col = "grey50", yaxt = "n", ylab = "",
     xlab = "Absolute standardised mean difference")
points(abs(lp$SMD_adj), y, pch = 17, col = "#2c7fb8")
segments(abs(lp$SMD_raw), y, abs(lp$SMD_adj), y, col = "grey80")
axis(2, at = y, labels = lp$variable, las = 1)
abline(v = 0.1, lty = 2)
legend("bottomright", legend = c("Raw", "Wave-adjusted"),
       pch = c(16, 17), col = c("grey50", "#2c7fb8"), bty = "n", cex = 0.8)
```

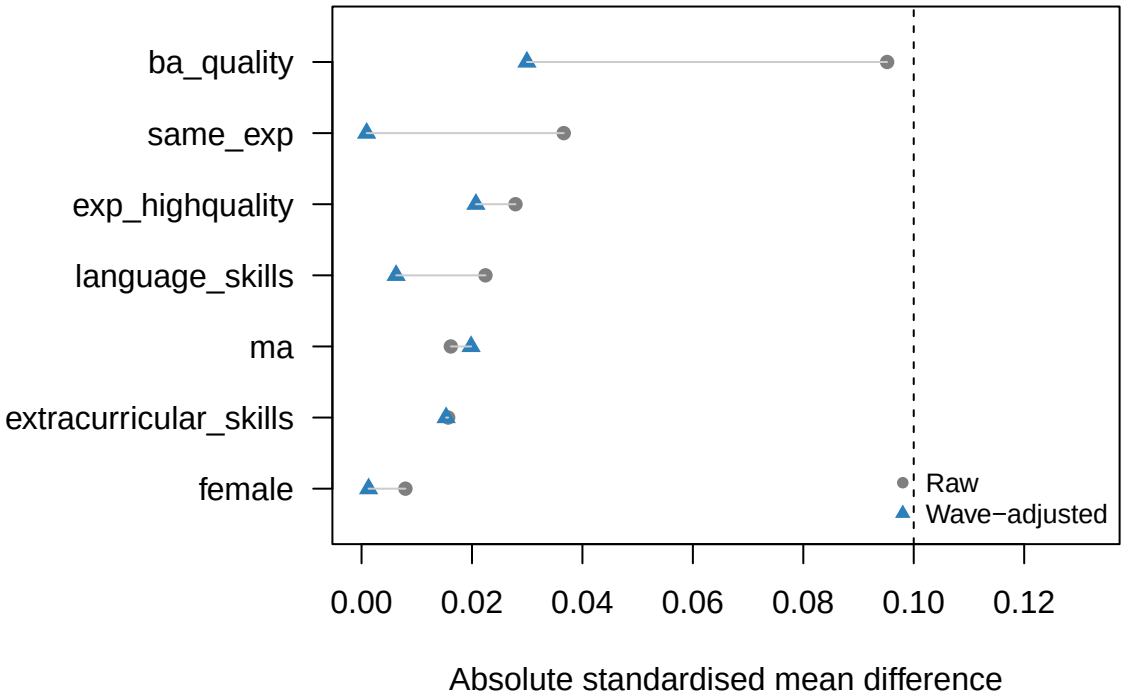


Figure 1: Absolute SMD before and after adjusting for wave (dashed line = 0.1 threshold)

```r
par(op)
```

4 Joint balance test

Another test we did to assess whether the restricted sample preserves covariate balance between treatment and control groups: We conducted a joint balance test by regressing treatment assignment on all pre-treatment resume characteristics and wave fixed effects:

$$D_i = \alpha + \sum_{j=1}^{k} \gamma_j X_{ij} + \delta_{w(i)} + \varepsilon_i$$

where D_i indicates treatment assignment (1 = foreign-sounding name, 0 = English-sounding name), X_{ij} represents the k pre-treatment resume characteristics, and $\delta_{w(i)}$ denotes wave fixed effects.

We test the joint null hypothesis $H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_k = 0$ using a Wald test with firm-clustered (CR2) standard errors. The Wald statistic is reported as an F-statistic ($F = \chi^2/k$), using the number of firms minus one as the denominator degrees of freedom. Wave fixed effects are included as controls but excluded from the joint restrictions.

A large p-value indicates insufficient evidence that the résumé characteristics jointly predict treatment assignment, conditional on wave. Since same_exp is only observed in Waves 0–1, we conduct two tests: one using all waves without same_exp, and another restricted to Waves 0–1 including same_exp.

```
joint_balance <- function(vars, label) {
  dat <- df_int1[complete.cases(df_int1[, vars]), ]
  fit <- lm_robust(as.formula(paste("D ~", paste(vars, collapse = " + "),
                                    "+ factor(fall_data)")),
                   data = dat, clusters = firmid, se_type = "CR2")
  b     <- coef(fit)[vars]
  V     <- vcov(fit)[vars, vars]
  wald  <- as.numeric(t(b) %*% solve(V) %*% b)
  k     <- length(vars)
  firms <- length(unique(dat$firmid))
  data.frame(sample  = label,
             resumes = nrow(dat),
             firms   = firms,
             k       = k,
             F_stat  = wald / k,
             p       = pf(wald / k, k, firms - 1, lower.tail = FALSE))
}

joint_tab <- bind_rows(
  joint_balance(setdiff(all_pretreat_cov, "same_exp"),
                "All waves, without same_exp"),
  joint_balance(all_pretreat_cov,
                "Waves 0-1, with same_exp")
)

kable(joint_tab, digits = 3,
      col.names = c("sample", "resumes", "firms", "covariates", "F", "p"),
      caption = "Joint balance test (D on all covariates + wave FE, firm-clustered)")
```

Table 4: Joint balance test (D on all covariates + wave FE, firm-clustered)

sample	resumes	firms	covariates	F	p
All waves, without same_exp	5657	3269	6	0.456	0.841
Waves 0-1, with same_exp	3397	1927	7	0.353	0.929

The joint tests fail to reject the null hypothesis that the included resume characteristics have no joint relationship with treatment assignment ($H_0 : \gamma_1 = \cdots = \gamma_k = 0$). For the full sample excluding same_exp, the test yields F = 0.456 (p = 0.841). Similarly, restricting the sample to Waves 0–1 and including same_exp yields F = 0.353 (p = 0.929).

Overall, the results are consistent with the original experimental randomization and provide no evidence that our sample restrictions introduced systematic imbalance in the observed resume characteristics between treatment and control groups, conditional on wave.